

Solution Architecture

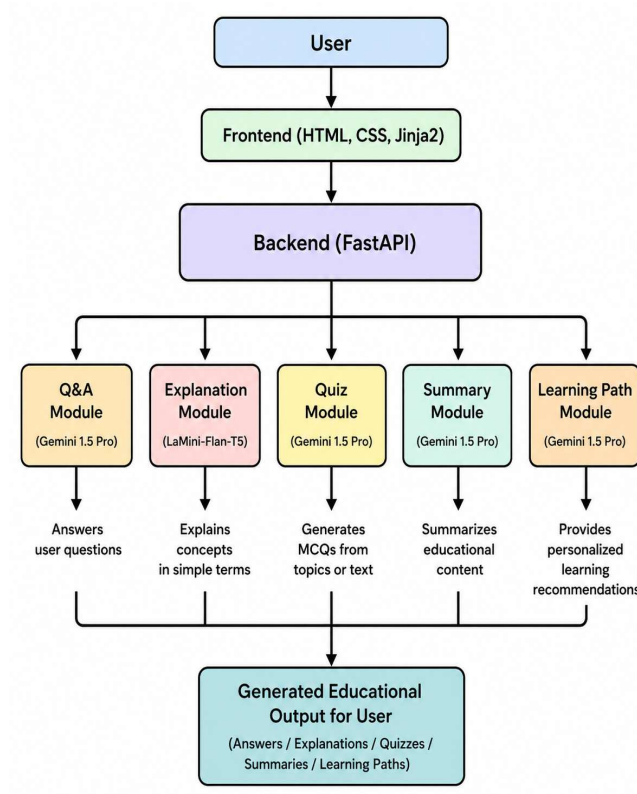
Date	15 March 2026
Team ID	xxxxxxx
Project Name	EduGenie – AI Learning Assistant
Maximum Marks	5 Marks

Solution Architecture Diagram:

EduGenie follows a modular three-tier architecture comprising Presentation Layer, Application Layer, and Data Layer.

Students interact with the platform through a responsive web dashboard.

The application processes requests through FastAPI services, communicates with Gemini AI, and stores educational records inside SQLite databases.



Architecture Layers

Presentation Layer

Student Dashboard

Authentication

Quiz Interface

Q&A Interface

Roadmap Builder

Summarizer

History Section

Analytics

Technology

HTML

CSS

JavaScript

Jinja2

API Gateway

FastAPI

JWT Validation

Authentication

Request Handling

Session Management

Core Logic Service

Gemini API

Quiz Engine

Summarizer

Q&A Engine

Recommendation System

Roadmap Generator

Analytics Service

External Services

Gemini API

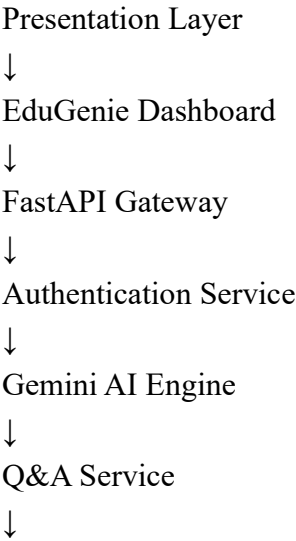
Data Layer

- SQLite
- User Profiles
- Learning History
- Quiz Records
- Summaries
- Roadmaps
- Analytics

Component Description Table

Component	Role	Technologies
Presentation Layer	User Interface	HTML CSS JS Jinja2
API Layer	Handles Requests	FastAPI
Authentication	Secure Access	JWT
AI Services	Educational Intelligence	Gemini API
Recommendation Engine	Personalized Suggestions	Python
Database	Data Storage	SQLite
Analytics	Learning Insights	Chart.js
History Module	User Records	SQLite

Architecture Diagram Labels



Quiz Generator



Summarizer



Roadmap Builder



SQLite Database



User Learning History



Analytics Dashboard

ER DIAGRAM - EDUGENIE LEARNING ASSISTANT

